

AAU 2K11/19 Meds Module 2 questions

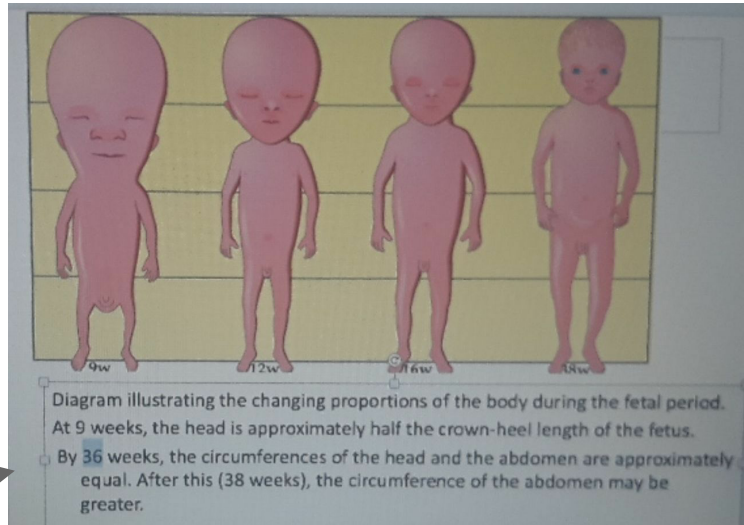
(includes remembered and repeated questions from previous years exams)

EMBRYOLOGY

1. Circumference of the abdomen equals circumference of the head at which week of fetal development?

- A. 16 weeks
- B. 36 weeks
- C. 9 weeks
- D. 38 weeks

ANS. B



2. There was a question about crus of diaphragm yemilew. Which one is true about diaphragm development?

The answer was Crura or crus develops from mesentery of esophagus

Thus the diaphragm is derived from the following structures:

- 1) the septum transversum, which forms the central tendon of the diaphragm;
- 2) the two pleuroperitoneal membranes;
- 3) muscular components from the lateral and dorsal body walls; and
- 4) the mesentery of the esophagus, in which the crura of the diaphragm develop

3. Umbilical cord entanglement is common in which type of twins?

- A. Monoamniotic, dichorionic
- B. Monoamnio, monochorionic
- C. Dichrionic, monoamniotic
- D. Monochrionic, diamniotic
- E. Diamniotic, dichorionic



Monocho-monoamn- This method of development is very uncommon. Division of the embryonic disc results in two embryos within one amniotic sac.

- A) Complete division of the embryonic disc gives rise to twins. Such twins rarely survive because their umbilical cords are often so **entangled that** interruption of the blood supply to the fetuses occurs.
- B) B and C) Incomplete division of the disc results in various types of conjoined twins.

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ANS. B

4. Condition in which fetal cords insert in the membranes covering the internal cervical os.

- A. Placenta previa
- B. Vas previa
- C....
- D....
- E....

velamentous cord insertion and vasa previa ...cont

- Because the vessels are attached to the chorion, rupture of the fetal membranes may rupture the vessels, which can result in fetal distress and death within minutes.
- If the fetal vessels insert in the membranes covering the internal cervical os = vasa previa.
- Because of the lack of protection from Wharton's jelly, these vessels are prone to compression and rupture, especially when they are located in the membranes covering the cervical os (ie, vasa previa).

ANS. B

5. A question about Fetal hydropsy and Erythroblastosis Fetalis

CLINICAL CORRELATES

Erythroblastosis Fetalis and Fetal Hydrops

- Antigens of the fetal RBC (Rh - antigens) can stimulate a maternal antibody response against fetal blood cells.
- The maternal antibodies will attack and hemolyze fetal red blood cells resulting in **hemolytic disease of the newborn (erythroblastosis fetalis)**.
- The anemia may become so severe that fetal hydrops (edema and effusions into the body cavities) occurs, leading to fetal death.

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- The antibody response occurs in cases where the **fetus is D(Rh) positive** and **the mother is D(Rh) negative** and is elicited when fetal red blood cells enter the maternal system due to small areas of bleeding at the surface of placental villi or at birth.
- The disease is prevented by identifying women at risk using an antibody screen and treating them with anti-D-immunoglobulin.

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*Read about these two conditions. In the above picture lay yalut sentences rasachew as choices metew neber. Tyakew which one is true neber milew, the answer was "All"

6. Which one is in correct about velamentous cord?

A. The cord can end several centimeters from the placenta

B. The umbilical vessels do not separate from each other and do not cross between the amnion and chorion.

C..... D..... E. All of the above

velamentous cord insertion and vasa previa

- The cord can end several centimeters from the placenta, at which point the **umbilical vessels separate from each other and cross between the amnion and chorion** before connecting to the subchorionic vessels of the placenta.

ANS. B,....I think "in correct" yalut "Incorrect" lemalet felegew new, Girmai or Brevis tyake siyawetu sew asdengituachew new accidentally space yechemerubet mehal lay.

7. There was a question about conjoined twins.

*Know this: conjoined twins are the result of incomplete division of the embryonic disc.

Focus on this part bedemb, there were many questions related to twins development (Multifetal pregnancy/ Gestation), especially about twin-twin transfusion syndrome,

- This syndrome occurs in as many as 30% of monochorionic-diamniotic MZ twins.
- There is shunting of arterial blood from one twin through arteriovenous anastomoses into the venous circulation of the other twin.
- **The donor twin** is small, pale, and anemic, whereas the **recipient twin** is large and polycythemic.
- The placenta shows similar abnormalities; the part of the placenta supplying the anemic twin is pale, whereas the part supplying the polycythemic twin is dark red.
- In lethal cases, death results from **anemia** in the donor twin and **CHF** in the recipient twin.

8. All of the following are clinical features that occur following congenital rubella syndrome Except

- A. Cataracts
- B. Cardiac defect
- C. Deafness
- D. Neural tube defect
- E. None

ANS. D

Infectious agents	
Rubella virus	Cataracts, glaucoma, heart defects, deafness, teeth defects
Cytomegalovirus	Microcephaly, blindness, mental retardation, sensorineural hearing loss, hepatosplenomegaly, fetal death
Herpes simplex virus	Microphthalmia, microcephaly, retinal dysplasia
Varicella virus	Limb hypoplasia, mental retardation, muscle atrophy
HIV	Microcephaly, growth retardation
Toxoplasmosis	Hydrocephalus, cerebral calcifications, microphthalmia
Syphilis	Mental retardation, deafness

9. Which of the following does not fit the properties of Endocrine cells

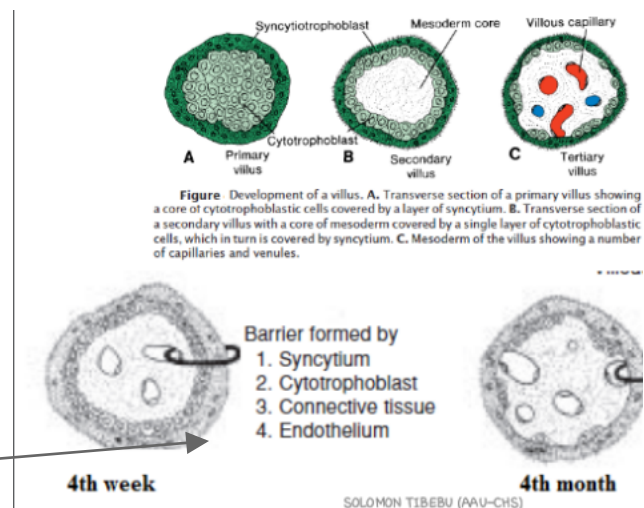
- A. Have similar manner of initial development as exocrine
- B. Cells arranged as clusters or follicles
- C. Have more blood capillaries
- D. Target large distance meselegn

ANS.....

10. The order of the placental barrier from inside to outside

- A. Connective tissue-cytotrophoblast-Syncytium-Endothelium
- B. Endothelium-Connective tissue-Cytotrophoblast-Syncytium
- C.....
- D.....
- E.....

ANS. B



11. Parietal layer covering pleura is derived from.

ANS. Somatic extraembryonic mesoderm.

12. Which one of the ff is wrong about Notochord?

- A. It induces the overlying ectoderm to... mnamn
- B. It extends from the primitive node to prechordal plate...
- C.....
- D.....
- E.....

ANS. Couldn't recall the other choices. A & B are true, # Read about the developmental process of the Notochord.

13. Which is true/false about mnamn....(it was hCG/human chorionic gonadotropin/ related question)

Choices..... It stimulates corpus luteum to grow to corpus albicans, hCG prevents corpus luteum from degenerating, Estrogen maintains the corpus luteum minamin..., Progesterone....

COMPLETION OF IMPLANTATION AND CONTINUATION OF EMBRYONIC DEVELOPMENT

- Implantation of the blastocyst completed by the end of 2nd week.
- As the blastocyst implants, more trophoblast contacts the endometrium and differentiates into
 - 1• The cytotrophoblast:
 - a layer of cells that is **mitotically active**
 - forms new cells that migrate into the increasing mass of syncytiotrophoblast, where they fuse and lose their cell membranes.
 - 2• The syncytiotrophoblast:
 - rapidly expanding, **multinucleated** mass in which no cell boundaries are discernible.
 - **erosive** - invades the endometrial CT, and the blastocyst slowly embeds itself in the endometrium.
 - displaces endometrial cells at the implantation site.
 - produces a hormone-**human chorionic gonadotrophin (hCG)**, which maintains the hormonal activity of the corpus luteum in the ovary during pregnancy.

- The endometrial cells undergo **apoptosis** (programmed cell death), which facilitates the invasion.
- The CT cells around the implantation site accumulate **glycogen and lipids** and assume a polyhedral appearance.
- Some of these cells-**decidual cells**-degenerate adjacent to the penetrating syncytiotrophoblast → engulfs these degenerating cells, providing a rich source of embryonic nutrition.
- **hCG** maintains the hormonal activity of the corpus luteum in the ovary during pregnancy.
 - corpus luteum: endocrine glandular structure, secretes **estrogen** and **progesterone** to maintain pregnancy.
- Highly sensitive radioimmunoassays are available for detecting hCG and pregnancy and forms the basis for **pregnancy tests**.
- Enough hCG is produced by syncytiotrophoblast at the end of the 2nd week to give a positive pregnancy test, even though the woman is probably unaware that she is pregnant.

14. Regarding cleavage, which statement is False?

- A. Results in the formation of increasingly small cells
- B. Occurs as the zygote passes down the uterine tube
- C. Begins when the male and female pronuclei contact each other
- D. Enhances the corpus luteum to enlarge to form corpus albicans
- E. Is a series of rapid mitotic divisions while the zygote is still surrounded by the zona pellucida

ANS. D

15. Beating of the heart can be heard using ultrasonography during theweek when counted from LNMP(last normal menstrual period)?

- A. 5 weeks
- B. 7 weeks
- C. 3 weeks
- D. 9 weeks
- E.....

ANS. B, 7 weeks.....from Fertilization period bihon nomo 5 weeks

16. Sacrococcygeal teratoma

- A. Remnants of primitive streak
- B. A benign tumor that contains tissues derived from all the three germ layers
- C. Most affected infants (80%) are females
- D. All of the above
- E. None of the above



Female infant with a large sacrococcygeal teratoma that developed from remnants of the primitive streak. The tumor, a neoplasm made up of several different types of tissue, was



Sacrococcygeal teratoma resulting from remnants of the primitive streak. These tumors may become malignant

ANS. D

17. The notochord

- A. Originates as solid cord of cells from the cranial end of the primitive node
- B. It persists in the adult only as nucleus pulposus of the intervertebral discs
- C. It induces the overlying ectoderm to thicken and form the neural plate
- D. Chordomas are tumors arising from vestigial remnants of notochordal tissue
- E. All of the above

ANS. E

18. During_____folding of the embryo, part of the yolk sac is incorporated into the embryo and become the midgut.

- A. Head
- B. Tail
- C. Lateral
- D. All
- E. None

ANS. C

19. Which of the ff structure does not turn under the ventral surface of the embryo during folding of the head?

- A. Prochordal plate
- C. Connecting stalk
- B. Heart
- D. Pericardial cavity
- E. Septum transversum

ANS. C

20. Before the folding of the embryo, the heart begins its development

- A. Extraembryonically in a region anterior to the prochordal plate
- B. Intraembryonically in a region ventral to the prochordal plate
- C. Intraembryonically in a region caudal to the prochordal plate
- D. In the intraembryonic mesoderm of the prochordal plate
- E. None of the above

ANS.....

21. Not derivative of neural crest cell

- A. Dorsal root ganglion of spinal nerve
- B. Neurolemma sheaths of peripheral nerves
- C. Adrenal medulla
- D. Autonomic ganglion
- E. None of the above

ANS. E

22. All of the ff epithelial linings/cells are derived from endoderm EXCEPT

- A. Epithelial lining of GIT, respiratory tract
- B. Epithelial cells of tonsils, thyroid gland
- C. Neuroepithelium of sense organs
- D. Epithelial lining of Eustachian tube
- E. None of the above

ANS. C

23. Most birth defects occur during which of the following periods/stages of the intrauterine life?

- A. First 2 weeks
- C. Fetal period
- B. Embryonic period
- D. Last trimester
- E. Zygote

ANS. B

24. Concerning multiple pregnancies,

- A. The rate of dizygotic twinning increases with maternal age.
- B. The frequency of monozygotic twinning shows marked racial differences.
- C. Twins with one amnion, one chorion and one placenta are always monozygotic.
- D. Dizygotic twins always have two amnions and two chorions.
- E. Monozygotic twins are of the same sex.

ANS. C

25. During the fetal period, which statement is true?

- A. Pupillary light reflex occurs during weeks 26-29.
- B. The amount of white fat is about 16% of body weight at 20 weeks.
- C. The spleen is the major site of erythropoiesis during the end of the 9th week.
- D. By 18 weeks the uterus is formed and canalization of the vagina is completed.
- E. The bodies of 12- week fetuses are usually completely covered with lanugo hairs

ANS. C

26. Intermediate medoderm gives rise to the

- A. Neural tube
- B. Heart
- C. Kidneys and gonads
- D. Somites
- E. Notochord

ANS. C

27. Which of the following is/are NOT mesodermal in origin?

- A. Microglia
- B. Bone and cartilage
- C. Astrocytes
- D. Cardiac muscle cells
- E. Endothelium of blood and lymph vessels

ANS. C

28. During the third week of development, which statement is false?

- A. Somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
- B. Notochord forms the primitive axis of the embryo around which the axial skeleton will form.
- C. The primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.
- D. Mesoderm exists between the ectoderm and endoderm everywhere.
- E. Neural tube is formed.

ANS. D

29. Select the correct event during the fetal period.

- A. Fetal allantois is an important site of erythropoiesis during week 20.
- B. The quantity of white fat is about 8% of body weight during week 21.
- C. By 36 weeks, the circumferences of the head and abdomen are approximately equal.
- D. The eyelids open at week 18.
- E. Umbilical herniation is a normal event during the 38th week.

ANS. C

30. What germ layer forms the melanocytes?

- A. Endoderm
- B. Neural crest
- C. Paraxial mesoderm
- D. Intermediate mesoderm
- E. Lateral plate mesoderm

ANS. B

31. Which structure is/are NOT visible during the fourth week?

- A. Closed neuropores
- B. Three pairs of pharyngeal arches
- C. Scalp vascular plexus
- D. upper limb buds
- E. Lens placodes.

ANS. C

32. During the embryonic period,

- A. Pigment is visible in the retina
- B. Sex of the embryo is clearly distinguishable externally
- C. Ossification of the fetal skeleton is active.
- D. The amount of white fat is about 16% of body weight.
- E. The body of the embryo is usually completely covered with lanugo hairs and vernix caseosa.

ANS. A

33. Select the incorrect statement.

- A. Gastrulation is first indicated by the formation of the primitive streak
- B. Neurulation is the process by which neuroectoderm forms the neural plate.
- C. Preponderance of sacrococcygeal teratoma is found in male newborns.
- D. All germ layers are derived from epiblast.
- E. Epiblast and hypoblast are fused in the cloacal membrane.

ANS. C

34. Which statement is false about prochordal plate?

- A. Helps to establish the bilateral symmetry of the embryo.
- B. Helps to establish the cranial versus the caudal region of the embryo.
- C. It eventually develops into oropharyngeal membrane at the site of the future mouth.
- D. It is columnar-shaped thickening of epiblast cells.
- E. It is a circular, midline thickening of hypoblast cells.

ANS. D

35. With regard to the notochord, which statement is false?

- A. It forms the nucleus pulposus of each intervertebral disk
- B. It is a solid cylinder of mesoderm extending from the primitive node to the prochordal plate.
- C. Provides signals that are necessary for the development the central nervous system
- D. It induces the formation of cardiogenic region.
- E. It induces the formation of each vertebral body

ANS. D

36. Which of the following structures must degenerate for blastocyst implantation to occur?

- A. Endometrium in progestational phase
- B. Zona pellucida
- C. Syncytiotrophoblast
- D. Cytotrophoblast
- E. Functional layer of the endometrium.

ANS. B

37. Where does the blastocyst normally implant?

- A. Functional layer of the cervix
- B. Myometrium
- C. Functional layer of the endometrium
- D. Perimetrium
- E. Basal layer of the endometrium

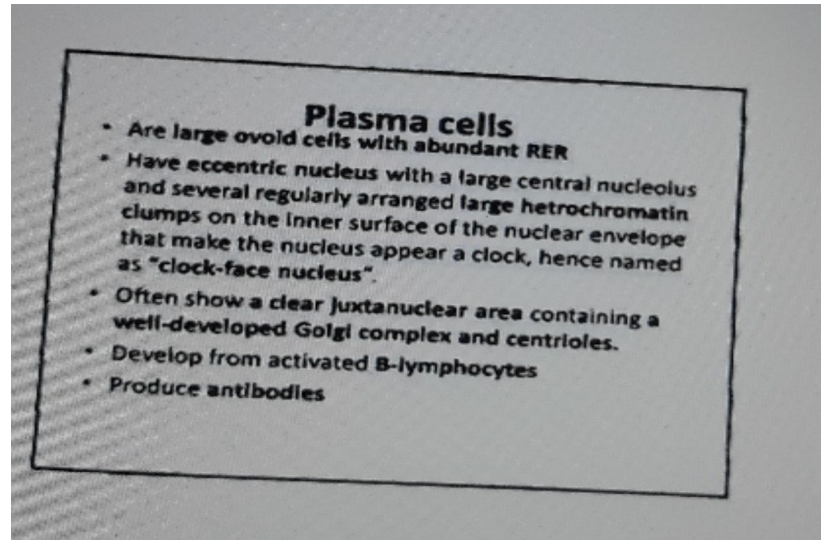
ANS. C

HISTOLOGY

38. Which connective tissue is derived from B lymphocytes and produces antibodies?

- A. Fibroblast
- B. Myofibroblast
- C. Histocyte
- D. Plasma cell
- E. Mast cell

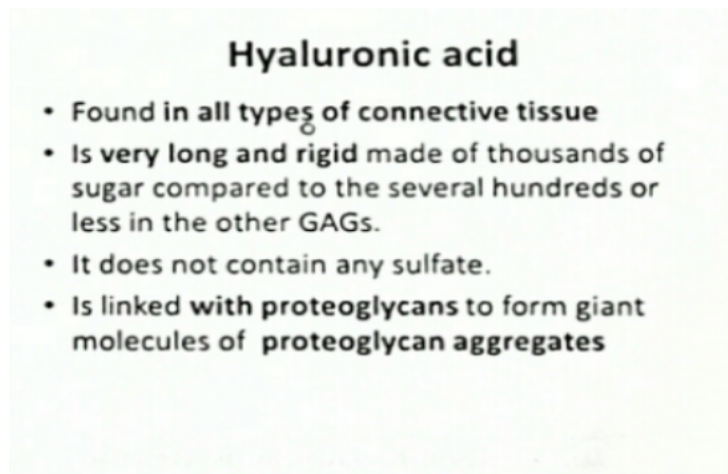
ANS. D



39. Glycosaminoglycan that is very long and rigid made of thousands of sugar and found in all types of CONNECTIVE TISSUE is

- A. Hyaluronic acid
- B. Chondroitin sulfate
- C. Dermatan sulfate
- D. Heparan sulfate
- E. Keratan sulfate

ANS. A



40. Regarding to the connective tissue fibers

- A. Elastic fibers make the stroma of many delicate organs
- B. Type IV collagen is the most abundant type of collagen fibers
- C. Type III makes a porous membrane in the basal lamina
- D. Collagen has high tensile strength with little elasticity
- E. Reticular fibers are composed of elastin and fibrillin

ANS. D

Collagen
• Found in all connective tissue • Is the most abundant protein of the body (about 30% of its dry weight) • Has high tensile strength with little elastic recoil

Tissue distribution of the most common types of collagen

Collagen Type	Tissue distribution	Synthesized by	Found as	Main function
I	Found in wide-spread tissue & accounts for 90% of body collagen	Various cells	Fibres & bundles	Resistance to force, tension & stretch
II	Hyaline & elastic cartilages, notochord & intervertebral disc	Chondrocytes	Fibrils	Resistance to intermittent pressure
III	Viscera, reticular laminae underlying basal laminae, around smooth muscle, endometrium, blood vessels & fetal skin	Various cells	Fibrils	Structural maintenance in expandable organs
IV	Basal laminae, kidney glomeruli & lens capsule	Various cells	Non-fibrillar network	Support & filtration barrier

this table is very important. Everything ሽምድድ!

41. Which cell to cell junction is the strongest?

- A. Zonula Occludens
- B. Zonula adherens
- C.....
- D.....
- E.....

ANS. NOT SURE, was a very አከራካሪ question, most probably gn B

42. Pseudounipolar has what kind of shape?

- A. Spherical
- B. Flat
- C.....
- D.....
- E.....

ANS. A

43. With regard to the connective tissue, which statement is false?

- A. Collagen is characterized by high tensile strength and flexibility.
- B. Plasma cells characteristically synthesize and secrete immunoglobulins.
- C. In ordinary connective tissue, the predominant component of ground substance is water.
- D. Fibroblast produces and maintains all extracellular matrix components of connective tissue.
- E. None of the above.

ANS. E

44. Concerning the connective tissue, which statement is false?

- A. Proteoglycans are responsible for the highly viscous character of the ground substance.
- B. In blood, the ground substance lacks stabilizing macromolecules.
- C. Reticular fibers are particularly abundant in the framework of hematopoietic organs.
- D. Macrophages are also antigen-presenting cells.
- E. None of the above.

ANS. E

***Histo read the slides, kal be kal keza lay new questions yewetut**

PHYSIOLOGY

45. Which of the ff has B1 receptor?

A. Brain B. GIT C. Lung D. Heart muscle E....

ANS. D (It was supposed to be Heart Muscle), The answer is heart muscle....the question lay gn spelling tezelo heat muscle new milew

Muscarinic	Adrenergic Receptors	
	$\alpha 1$ (Blood vessels)	$\alpha 1$ (Salivary)
M1 (Brain)	$\alpha 2$ (GIT)	$\alpha 2$ (Pancreas)
M2 (Heart)	$\beta 1$ (Heart)	$\beta 1$ (JGC)
M3 Smooth m. Parietal cells Salivary glands	$\beta 2$ (Lungs)	$\beta 2$ (GIT, Liver)
M4 (Smooth m. Pancreas)	$\beta 3$ (Adipose, $\alpha 1$, $\beta 1$) stimulate lipolysis	
M5 (Presynaptic modulation)		

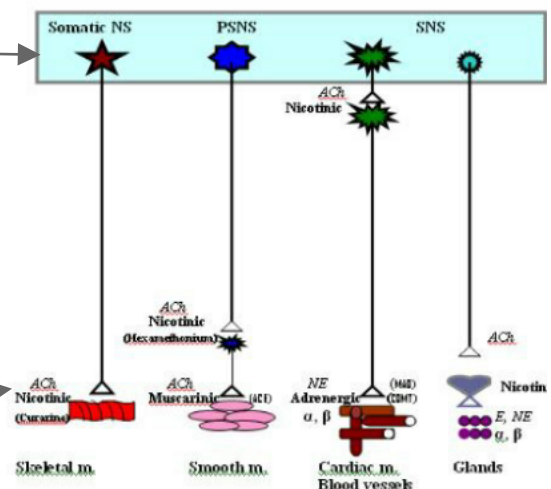
46. In somatic nervous system, the neuron releases Acetylcholine (ACh) to elicit nicotinic receptor found in...

A. Post synaptic membrane of the skeletal muscle

B..... C..... D...

E....

ANS. A



47. A question about muscarinic receptor

Choice wist skeletal muscle contraction neber (Not the answer though)

Muscarinic acetylcholine receptors

- On smooth muscles
- On cardiac muscle
- On gland cells
 - Nasal, lacrimal, parotid, gastric, pancreas, submandibular
- On some CNS neurons
- On some neurons of autonomic ganglia
 - **mAChR** - metabotropic-G protein coupled receptor which produce **slow EPSP** or **IPSP**
 - **nAChR** - produce fast EPSP

48. True about Myelinated neurons

A. AP propagation is fast than unmyelinated neurons

B.... C..... D..... E..... (Couldn't recall the other choices)

ANS. A

49. Which of the ff has huge role for resting membrane potential?

A. leakage channels for sodium

B. Leakage channels for potassium

C. Leakage channels for chloride D..... E....

ANS. B,..... For resting membrane potential huge role yalachew are leakage channels for potassium.

50. There is another question that says based on what new "yehone membrane" miyasalifew....

A. Osmolarity and size C..... D.....

B. Na and K concentration E.....

ANS. B

51. Which one is a Non-excitable cell?

A. Skeletal muscle C. Cardiac muscle E. Adipose

B. Smooth muscle D. Neuron F. Beta cell

ANS. E(Adipose/fat),.....Excitable cells include nerve and all muscle cells. Beta cell mitlewa taterateralech.

52. A non-diffusible ion inside a cell can affect the diffusible ions across the membrane...mnamn which can be proved by mathematical equation of.....

A. Gibbs Donnan Equilibrium

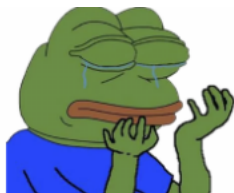
C. Goldman-katz Equation

B. Nernst equation

D.....

E.....

ANS.



Google

53. Which one of the ff is NOT Parasympathetic effect of ANS?

A. Liver-Glycolysis

B. Glycogenesis

C. Lipogenesis

D. Miosis

E.....

Effects of ANS	
PSNS	SNS
• Rest - Digest • Energy storage • Glycogenesis/none • Lipogenesis/none • Better GIT activity • Copious watery saliva • Biliary flow stimulated • More saliva/ Lacrimal/nasal/ pancreatic/gastric secretion • Visual miosis • Splanchnic vasodilation • Cardiac inhibition • Vasodilation (ABP)/none • Airway constriction • More blood flow to genitalia • Reduced adrenaline	• Fight-flight, alarm, stress • Energy expenditure • Glycogenolysis • Lipolysis • Reduce GIT activity • Thick saliva • Less saliva/lacrimal/pancreas secretion • Inhibit biliary secretion • Visual mydriasis • Skeletal muscle blood flow • Cardiac stimulation • Vasoconstriction (ABP) • Airway dilatation • Less blood flow to genitalia • Increases adrenaline • Odeiferous sweat

ANS. A,..... During parasympathetic meregagat silhone energy should be store through the processes of glycogenesis in the liver and lipogenesis, Glycolysis doesn't occur during Parasympathetic conditions

54. Demyelination of neurons in the CNS results in

A. Multiple sclerosis

C.....

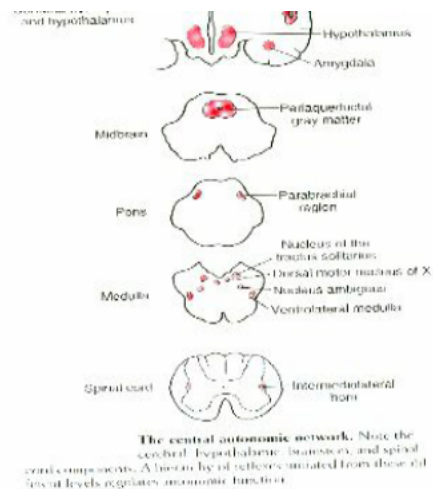
D.....

B. Guillain-Barre syndrome

E.....

ANS. A,.....Multiple sclerosis results from demyelination of neurons in CNS, while Guillain-Barre syndrome results from demyelination of neurons in PNS.

There were two questions on these, study them betam



• At the level of the Brain stem

- Vital centers

- **Periaqueductal gray matter (midbrain)** – autonomic response to pain
- **Ventrolateral medulla**
 - Contribute to the control rhythm of breathing
- **Locus ceruleus** – Micturition
- **Nucleus tractus solitarius** – sensory autonomic component. This integrates sensory information from the internal organs and coordinates output of CNS and brain stem

• At the level of the spinal cord

- Coordinate many autonomic reflexes

- **Vascular tone** (Episode of hypertension and piloerection after spinal cord injury is example of un inhibited SNS reflex)
- **Micturition**
- **Defecation**
- **Sexual responses**
- Higher centers are required to facilitate or inhibit the reflex centers

• Nucleus ambiguus (cardioinhibitory)

- Contain preganglionic
 - motor nuclei of vagus to the heart
 - Motor nuclei to soft palate, larynx, pharynx for speech and swallowing

• Dorsal motor nucleus of the vagus

- Contain preganglionic vagal fibers to GIT, lungs, abdomen, and thorax

• Parabrachial nucleus

- Contribute to respiration, BP response to hemorrhage and hypovolemia, taste, temperature, arousal state and glucose homeostasis

- Medullary centers control autonomic reflexes for coughing, sneezing, swallowing, vomiting, salivation, lacrimation.

- At the level of midbrain, control of
 - Control pupillary light reflex (Pretectal + EWN)
 - Accommodation reflexes (LGB+Visual cortex)
 - Corneal reflex (superior colliculus+VII)

- Responsible for autonomic changes that allows biological clock (suprachiasmatic nuclei)
- Lateral, dorsomedial & posterior periventricular area of hypothalamus are considered as sympathetic center
- (Hypothalamus is said to be the head ganglion of ANS)

66

67

- At the level of hypothalamus,
 - Receives about the body's status, anticipates some of its needs, provides coordinated set of neural and hormonal outputs
 - Control of body temperature, water balance, food intake, adrenaline secretion
 - The main regulator of autonomic preganglionic fibers in response to body status
 - Paraventricular zone is connected to brain stem and spinal nuclei to control ANS

Cerebral Cortex-Hypothalamus

- Insular cortex and medial prefrontal area contain sensory and motor area involved in regulation of ANS
- Unpleasant emotions coupled to generalized SNS discharge involves amygdala in temporal lobe
 - Heart rate, blood pressure, pupillary dilatation, adrenaline secretion
- Cerebral cortex-hypothalamus-locus coeruleus sometimes activates PSNS-micturition

68

55. Accommodation reflex occurs in

A. Medulla B. Midbrain C. Hypothalamus D. Hindbrain E. Spinal cord

ANS. B,...refer from the above pictures, very important!

56. What is the role of parasympatheticon....minamin

A. reduces coronary blood flow C. Increases blood flow to skeletal muscles
 B.Heart inhibition D..... E...

ANS. A,.. Couldn't recall the question mulu le mulu but the answer was A.

57. Which one of the ff are responsible for action potential?

A. Voltage gated Sodium channels

B.....

C.....

D.....

E.....

ANS. A

58. A question about neurons doctrine. Which is false about the neuron doctrine?

To answer this question,

know these

Neuron doctrine

- Individual neurons are the elementary **signaling elements** of the nervous system
- Electrical signals within the nerve cells flow only in one direction (**Principle of dynamic polarization**)
- Nerve cells make specific connections and not random or indiscriminate (**Principle of Connectional specificity**)
- There is **no cytoplasmic continuity** between two communicating neurons

* All the four sentences in the picture metew neber as choices, hulum true (also the second sentence was tinish modified..it says "nerve cells make discriminate connections", which is still true.) read them well!. They added another choice (a fifth sentence) which was false, the answer.

59. Which one of the ff is wrong

A. post ganglionic neurons of sympathetic are cholinergic(ACh)

B. Adrenergic sweat gland

C. Cholinergic.....

D.....

E....

ANS. A,....because they are not cholinergic rather adrenergic(they release norepinephrine not acetylcholine), Gn remember postsynaptic neurons for parasympathetic as well as all preganglionic neurons are cholinergic. Note the difference!

60. How is the sympatho adrenal axis elicited?

ANS. through the release of adrenaline.(Not sure, google it)

61. In which region of a neuron are voltage gated Na channels located?

A. Presynaptic membrane

C. Dendrites

B. Axon Hillock

D.....

E.....

ANS. B

ANATOMY

THE BACK

(62-65, from our tests)

(66-80, from Appleton)

62. Which of the following is the location of nucleus pulposus herniation on the vertebral column

ANS. Posterolaterally

63. Spina bifida is the result of defect in closure of which vertebrae

ANS. L5 and/or S1

64. Which nerve root is compressed when there is an L4-L5 disc herniation

ANS. L5 nerve

65. Where do u perform lumbar spinal puncture

ANS. L3 & L4 or L4 & L5

66. Which one of the ff muscles forms the floor of the lumbar triangle of the back region?

- A. External oblique
- B. Internal oblique
- C. Quadratus lumborum
- D. Transverse abdominis
- E. Latissimus dorsi

ANS. B

Muscular Triangles of the Back

- **Lumbar Triangle**
 - It may be the site of
 - ✓ an abdominal hernia and
 - ✓ pus may emerge from the abdominal wall.
 - **Boundaries**
 - ✓ **Posterior:** latissimus dorsi,
 - ✓ **Anterior:** Posterior border of the external oblique muscle of the abdomen, and
 - ✓ **Inferior:** Iliac crest
 - ✓ **Floor:** Internal oblique muscle

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67. The vertebral column does all of the following EXCEPT

- A. protect the spinal cord and spinal nerves
- B. support the weight of the body
- C. provide a pivot for the head
- D. play an important role in posture and locomotion
- E. form the main part of the appendicular skeleton

ANS. E,.....The vertebral column forms the main part of the axial skeleton

68. Which of the ff muscles of the back is NOT innervated by the dorsal rami of spinal nerves?

- | | |
|--------------------------|--------------------------------|
| A. Iliocostalis lumborum | D. Splenius capitis |
| B. Longissimus thoracis | E. Serratus posterior superior |
| C. Multifidus | |

ANS. E

69. Which of the following is NOT a distinctive characteristic of a typical cervical vertebra?

- A. The body is small and wider from side to side than anteroposteriorly.
- B. The vertebral foramen is large and triangular.
- C. The transverse processes contain transverse foramina.
- D. The articular processes contain superior facets directed inferoanteriorly.
- E. The spinous processes are short and bifid.

ANS. D,.....The articular processes contain superior facets directed superoposteriorly

X-Cs of Typical Cervical Vertebrae

- Transverse processes possess a foramen transversarium for the passage of the vertebral artery and veins
→ note that the vertebral artery passes through the transverse processes C1 to 6 and not through C7
- The spines are small (short) and bifid
- The body is small and broad from side to side
➢ has a concave superior surface & a convex inferior surface
- The vertebral foramen is large and triangular
- Articular processes
 - ◆ Superior facets - face backward and upward;
 - ◆ Inferior facets are face downward and forward.

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70. Which of the following is NOT a distinctive characteristic of a typical thoracic vertebra?

- A. The body is heart-shaped.
- B. The vertebral foramen is triangular and larger than in cervical and lumbar vertebrae.
- C. The transverse processes are long and strong and extend posterolaterally.
- D. The articular processes contain superior facets directed posteriorly and slightly laterally.
- E. The spinous processes are long and slope posteroinferiorly.

ANS. B,..... The vertebral foramen is circular and smaller than the foramina of cervical and lumbar vertebrae

Thoracic Vertebrae

- The middle four thoracic vertebrae (T5 - T8) demonstrate all the features typical of thoracic vertebrae
 - Vertebral body
 - ✓ Heart shaped; costal facets for articulation with head of rib
 - Vertebral foramen
 - ✓ Circular and smaller than those of cervical and lumbar vertebrae
 - Transverse processes
 - ✓ Long and strong and extend posterolaterally;
 - ✓ length diminishes from T1 to T12
 - ✓ T1-T10 have facets for articulation with tubercle of rib

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71. Which of the following is a distinctive characteristic of a typical lumbar vertebra?

- A. The body is massive and kidney-shaped when viewed superiorly.
- B. The vertebral foramen is circular and smaller than those of cervical and lumbar vertebrae.
- C. The transverse processes are long and slender and contain mammillary processes.
- D. The articular processes contain accessory processes.
- E. The spinous processes are long and slope posteroinferiorly.

ANS. A,.....The body is massive and kidney-shaped when viewed superiorly. The vertebral foramen is triangular and larger than the foramina in thoracic vertebrae and smaller than those in cervical vertebrae. The transverse processes are long and slender and contain accessory processes. The articular processes contain mammillary processes. The spinous processes are short, thick, and broad.

Lumbar Vertebrae

- Lumbar vertebrae are in the lower back between the thorax and sacrum
- The vertebral bodies are massive and progressively increase in size from vertebra L1 to L5
- Note that the lumbar vertebrae have no facets for articulation with ribs and no foramina in the transverse processes
 - **Vertebral body**
 - ✓ Massive; kidney shaped when viewed superiorly
 - **Vertebral foramen**
 - ✓ Triangular; larger than in thoracic vertebrae and smaller than in cervical vertebrae

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Lumbar Vertebrae

- **Transverse processes**
 - ✓ Long and slender;
 - ✓ **Accessory process** on posterior surface of base of each process; provides an attachment for the **intertransversarii muscles**.
- **Articular processes**
 - ✓ Nearly vertical facets;
 - **Superior facets** directed posteromedially (or medially);
 - has **mammillary process**; give attachment to both multifidus & intertransversarii muscles of the back.
 - **Inferior facets** directed anterolaterally (or laterally);
- **Spinous process**
 - ✓ Short, flat, and quadrangular and project backward

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72. Which of the following is true regarding the intervertebral disc between the C1 and C2 vertebrae?

- A. Its annulus fibrosus is composed of concentric lamellae of fibrocartilage.
- B. It does not contain a nucleus pulposus as other intervertebral discs do.
- C. It is thicker than other intervertebral discs.
- D. It acts like a shock absorber in response to axial forces.
- E. There is no intervertebral disc between the C1 and C2 vertebrae.

ANS. E,.....There is no intervertebral disc between the C1 and C2 vertebrae

73. Spinal arteries supplying the vertebrae are branches of the

- A. vertebral and ascending cervical arteries in the neck
- B. posterior intercostal arteries in the lumbar region
- C. subcostal and lumbar arteries in the pelvis
- D. iliolumbar and lateral and medial sacral arteries in the thorax
- E. aorta

Vasculature of the Vertebral Column

Arteries includes

Neck region	Thoracic region	Abdominal region	Pelvic region
▪ Vertebral ▪ Ascending cervical	▪ Posterior intercostal	▪ Subcostal ▪ Lumbar	▪ Iliolumbar ▪ Lateral sacral ▪ Medial sacral

ANS. A,..... Spinal arteries supplying the vertebrae are branches of the vertebral and ascending cervical arteries in the neck, posterior intercostal arteries in the thorax, subcostal and lumbar arteries in the lumbar region, and iliolumbar and lateral and medial sacral arteries in the pelvis

74. Which of the following back muscles is innervated by dorsal rami?

- A. latissimus dorsi C. rhomboid major
- B. levator scapulae D. rhomboid minor E. longissimus

ANS. E,.....The longissimus, a deep or intrinsic back muscle, is innervated by dorsal rami. All superficial or extrinsic back muscles except the trapezius, which is innervated by the accessory nerve, are innervated by ventral rami

75. Which of the following is NOT a deep (or intrinsic) muscle of the back?

- A. serratus posterior inferior D. multifidus
- B. levatores costarum E. splenius capitis
- C. iliocostalis

ANS. A,..... The serratus posterior inferior is an intermediate extrinsic back muscle

76. The splenius capitis and cervicis

- A. extend the head and neck
- B. flex the head and neck
- C. elevate ribs, assisting inspiration
- D. stabilize the atlas and axis
- E. move the upper limb

ANS. A,..... The splenius capitis and cervicis, when acting together, extend the head and neck

77. The erector spinae muscles

- A. flex the vertebral column D. prevent lateral bending of the vertebral column
- B. flex the head E. assist with elevation
- C. control flexion of the back

ANS. C,.....The erector spinae muscles extend the vertebral column and head when acting bilaterally, control flexion of the back by gradually lengthening their fibers, and bend the vertebral column laterally when acting unilaterally

78. Which of the following muscles does NOT attach to transverse processes of vertebrae?

- A. semispinalis B. multifidus C. rotatores D. intertransversarii E. spinalis

ANS. E,.....The spinalis inserts on spinous processes not transverse processes. The semispinalis, multifidus, rotatores, and intertransversarii all attach to transverse processes of vertebrae

79. The transversospinalis muscles do all of the following EXCEPT

- A. extend the head
- B. extend the thoracic and cervical regions of the vertebral column
- C. elevate ribs, assisting inspiration
- D. stabilize vertebrae
- E. assist with rotation of the vertebral column

ANS. C,.....The transversospinalis muscles do not elevate the ribs to assist inspiration

80. Which of the following muscles does NOT laterally bend the cervical intervertebral joints?

- A. longus colli
- B. iliocostalis cervicis
- C. longissimus capitis and cervicis
- D. splenius capitis
- E. splenius cervicis

ANS. A,.....The longus colli flexes the cervical inter vertebral joints but does not bend them laterally

UPPER LIMB

81. Most commonly the elbow dislocates_____

- A. Anteriorly B. Posteriorly C..... D..... E.....

ANS. B

82. Innervation to the rotator cuff muscle that laterally rotates the arm is provided by which of the ff?

- A. Suprascapular nerve
B. Median nerve
C. Ulnar nerve
D. Upper and lower subscapular nerves
E. Dorsal scapular nerve

E. The Teres Minor Muscle

- This is an elongated, tapering muscle and is often inseparable from the infraspinous muscle, which lies along its superior border,
Proximal attachment: superior part of lateral border of scapula,
Distal attachment: inferior facet on greater tubercle of humerus,
Innervation: axillary nerve (C5 and C6)
Action: the teres minor laterally rotates the arm and assists in its adduction. It also helps the other muscles of the rotator cuff to stabilise the shoulder joint.

F. The Infraspinatus Muscle

- This is a triangular muscle that occupies most of the infraspinous fossa,
Proximal attachment: infraspinous fossa of scapula,
Distal attachment: middle facet on greater tubercle of humerus,
Innervation: suprascapular nerve (C5 and C6)

ANS. A,.... it can be either axillary nerve (for Teres minor) or suprascapular nerve (for infraspinatus)

83. Cutaneous nerve supplying the skin covering the radial part of three and half digits

- A. Recurrent branch of median nerve
B. Lateral branch of median nerve
C. Medial branch of median nerve
D. Palmar cutaneous nerve of median nerve
E. None of the above

ANS. D

84. Which of the ff is wrong about structures association and their location

- A. Quadrangular space - Circumflex subscapular artery
- B. Triangular interval - Profunda brachii artery
- C. Deep branch of radial nerve - between heads of supinator muscle
- D. Ulnar nerve - anterior to triceps muscle in upper half of arm
- E. None

ANS. A,....through quadrangular space myalfut posterior circumflex humeral artery and axillary nerve nachew

Anatomical spaces on the back of the shoulder

1. Quadrangular Space: is a potential intermuscular with boundaries

Superior: subscapularis in front and teres minor behind and in between these two

muscles is the capsule of the shoulder joint

Inferior: teres major **Medial:** long head of triceps **lateral:** surgical neck of humerus

- o Axillary nerve and posterior circumflex humeral vessels pass through this space

2. Triangular space:

Superior: teres minor **Inferior:** teres major **Lateral:** long head of triceps

- o Circumflex scapular artery passes through this space to the infraspinous fossa

❖ **The triangular interval :**

Superior: teres major **Medial:** long head of triceps **lateral:** lateral head of triceps

- o radial nerve and profunda brachii vessels pass through this interval

85. The rotator cuff muscles originate from all of the ff surface of scapulae EXCEPT

- A. Subscapular fossa
- B. Supraspinatus fossa
- C. Infraspinatus fossa
- D. Lateral border of scapulae
- E. Inferior angle of scapulae

ANS. E

86. Which of the ff muscle of the upper limb originates from three different places?

- A. Extensor digitorum
- B. Triceps
- C. Flexor digitorum superficialis
- D. Flexor digitorum profunda
- E. Abductor digiti minimi

ANS. B, triceps has three origins: long head, medial head & lateral head

87. The surface marking in the scapulae serve for attachment (insertion) of the serratus anterior axioapendicula muscles is

- A. Outer surface of medial border of scapulae
- B. Superior border of the scapulae
- C. Lateral border of the scapulae
- D. Inferior angle of the scapulae
- E. None of the above

ANS. A

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c. The Serratus Anterior Muscle

- This is a large, foliate muscle that overlies the lateral portion of the **thorax** and the **intercostal muscles**.
- It was given its name (L. serratus, a saw) because the **saw-toothed appearance** of the fleshy digitations at its proximal attachments.
ORIGIN: external surface of lateral parts of ribs 1 to 8.
INSERTION: anterior surface of medial border of scapula.
INNERVATION: long thoracic nerve (C5, C6, and C7).
ACTION : protracts the scapula and holds or fixes it against the thoracic wall.
 ----Because it is active during punching, it has been called "**the boxer's muscle**".
 ---- By fixing the **scapula** to the thorax, it acts as an **anchor** for this bone and permits other muscles to use it as a fixed bone to produce movements of the humerus.
 -----Inferior fibres of Serratus anterior help to **raise the glenoid fossa** of the scapula.
- "**winged scapula**": paralysis of serratus anterior due to long thoracic nerve injury.

90. The posterior relation of the first part of the axillary artery is

- A. Long thoracic nerve
- B. Chords of brachial plexus
- C. Axillary vein
- D. Posterior cord
- E. None

A, First Part of the Axillary Artery

- This part is enclosed in the axillary sheath along with the axillary vein.
- Relations: pectoralis major and cephalic vein (anteriorly), long thoracic nerve (posteriorly), cords of the brachial plexus (laterally), and axillary vein (medially).
- This part of the axillary artery has only one branch, the supreme thoracic artery.
- This artery helps to supply the first two intercostal spaces and the superior part of the serratus anterior muscle.

ANS. A,....

91. The ff tendons are inserted into the base of the proximal phalanx of the thumb except

- A. Extensor pollicis brevis
- B. Abductor pollicis longus
- C. Oblique head of adductor pollicis
- D. Flexor pollicis brevis
- E. Transverse head of adductor pollicis

ANS. B,.....Abductor pollicis is inserted into the base of the first metacarpal NOT into proximal phalanx of the thumb

92. A muscle with a single origin

- | | |
|-----------------------------------|-------------------------|
| A. Pronator teres | D. Flexor carpi ulnaris |
| B. Flexor digitorum superficialis | E. None |
| C. Adductor pollicis | |

ANS. E

93. Which of the ff artery doesn't form the arterial anastomosis around the scapula

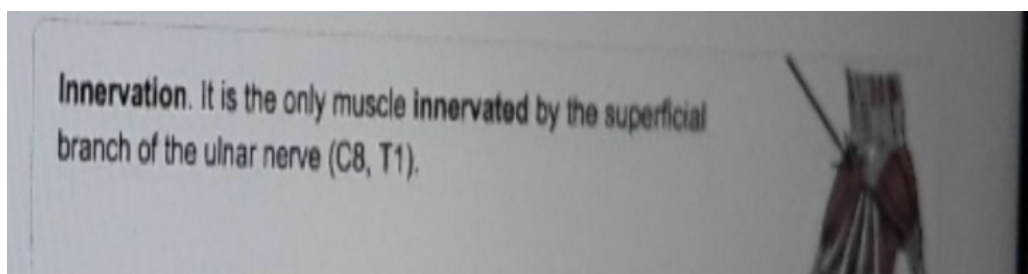
- A. Anterior circumflex humeral artery
- B. Profunda brachial artery
- C. Suprascapular artery
- D. Thoracoacromial artery
- E. Transverse cervical artery

ANS. B

94. Which of the ff is innervated by superficial branch of ulnar nerve?

- A. Flexor pollicis longus
- B. Extensor pollicis longus
- C. Abductor pollicis brevis
- D. Deep head of flexor pollicis brevis
- E. Palmaris brevis

ANS. E,..... Palmaris brevis is a hand muscle and it originates from where the palmaris longus inserts, that's from the palmar aponeurosis and flexor retinaculum and it inserts on the palm on the hypothenar side of the hand. It's the only muscle innervated by the superficial branch of the ulnar nerve.



95. Structure on the anterior aspect of the wrist beneath the flexor retinaculum include all EXCEPT

- | | |
|-----------------------------------|---------------------------|
| A. Palmaris longus | D. Flexor pollicis longus |
| B. Flexor digitorum profunda | E. Median nerve |
| C. Flexor digitorum superficialis | |

ANS. A,.....Palmaris longus passes superficially above the flexor retinaculum.

96. Loss of supination of the flexed forearm suggests a lesion in the

- | | |
|---------------------------|-------------------|
| A. Radial nerve | D. Median nerve |
| B. Musculocutaneous nerve | E. Axillary nerve |
| C. Ulnar nerve | |

ANS. Bmusculocutaneous nerve innervates biceps ...biceps demo is a powerful supinator when the forearm is flexed. But prime mover in supination of the forearm and hand is the supinator muscle. The pronator quadratus initiates pronation and is assisted by the pronator teres when more speed and power are needed.

97. The intrinsic muscle of the hand having dual innervation is

- | | |
|---------------------------|---------------------------|
| A. Flexor pollicis brevis | D. Abductor digiti minimi |
| B. Adductor pollicis | E. Opponens digiti minimi |
| C. Opponens pollicis | |

ANS. A

98. All except one of the ff muscle is supplied by two nerves

- A. Brachialis
- C. Subscapularis
- B. Flexor pollicis brevis
- D. Flexor digitorum profunda
- E. None

ANS. E

99. The arterial supply of the thumb is derived from

- A. Common palmar digital artery
- B. Superficial palmar arch
- C. Deep palmar arch
- D. Princeps pollicis artery
- E. Superficial branch of radial artery

ANS. D,.....for the index finger bñon noro demo radialis indicis neber mihonew. These two (Princeps pollicis artery and radialis indicis artery) are branches of the radial artery.

100. One of the ff branch of median nerve arise before the nerve crosses the flexor retinaculum

- A. Recurrent branch of median nerve
- B. Lateral branch of median nerve
- C. Medial branch of median nerve
- D. Palmar cutaneous branch of median nerve
- E. None

ANS. D (Not sure!)

101. The triangle of auscultation is bordered by all of the ff EXCEPT

- A. The superior horizontal border of the latissimus dorsi
- B. The medial border of scapula
- C. The interolateral border of trapezius
- D. The lateral border of scapula
- E. None

ANS. D

102. The sensory innervation of the nail bed of the index finger is the

- A. Median nerve
- B. Radial nerve
- C. Dorsal cutaneous branch of the ulnar nerve
- D. Superficial branch of the ulnar nerve
- E. Palmar cutaneous branch of the ulnar nerve

ANS. A

103. A question about lymphatic system of the female breast, lateral part of the breast is drained by which group of lymph nodes?

Lymphatic Drainage of the Breast

- Most of the lymphatic drainage (about 75%) is to the axillary lymph nodes, mainly the pectoral group.
- Lymph from the medial part of the breast drains into the parasternal lymph nodes, which are located within the thorax along the internal thoracic vessels.

NN; 2-6th intercostals and supraclavicular nerves.

104. One of the ff muscle doesn't originate from the lateral border of scapula

- | | | |
|----------------|---------------------|--------|
| A. Teres minor | C. Latissimus dorsi | |
| B. Teres major | D. Triceps brachii | E..... |

ANS. C

105. Which of the ff muscle is NOT from superficial flexor group of muscles of the forearm?

- | | |
|---------------------------|-----------------------------------|
| A. Flexor carpi radialis | D. Flexor digitorum superficialis |
| B. Flexor pollicis longus | E. Palmaris longus |
| C. Pronator teres | |

ANS. B.....but somehow D'm lihon yechlal intermediate selehone.

106. Which of the ff artery/arteries is/are branch of deep palmar arch?

- A. Proper digital arteries
- B. Princeps pollicis artery
- C. Common palmar digital arteries
- D. Palmar metacarpal arteries
- E. Deep palmar artery

ANS. D,..... Common palmar digital arteries are from superficial arch while palmar metacarpal arteries(they are 3 in number) are from deep palmar arch

107. Which one doesn't pass through minamin milu questions neberu.

Know these things

*Ulnar nerve and ulnar artery don't pass through the carpal tunnel or don't pass through the flexor retinaculum, they pass through Guyon's canal

*Structures passing under flexor retinaculum (yaw same us be carpal tunnel yemialfu)

*Structures passing through extensor retinaculum

*Structures passing through cubital fossa.

*About anatomical snuffbox too

* The six synovial sheaths. There are 6 compartments or synovial sheaths under extensor retinaculum.....

1st. Tendons of Abductor pollicis longus and extensor pollicis brevis

2nd. Tendons of extensor carpi radialis brevis and extensor carpi radialis longus

3rd. Tendon of extensor pollicis longus

4th. Tendons of extensor digitorum and extensor indicis

5th. Tendon of digiti minimi

6th. Tendon of extensor carpi ulnaris

The Extensor Retinaculum

- attached by septa to the radius and medial carpals (pisiform & triquetrum), having tunnels

Extensor tunnels:- 6 synovial sheaths- From lateral to medial

1. Abductor pollicis longus and extensor pollicis brevis
2. Extensor carpi radialis longus and brevis
3. Extensor pollicis longus
4. Extensor digitorum and extensor indicis
5. Extensor digiti minimi
6. Extensor carpi ulnaris



108. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper, when you try to pull it from the fingers, and the patient is unable to perform this action, which nerve is suspected to be non-functional

- A. Deep branch of radial nerve
- B. Superficial branch of ulnar nerve
- C. Recurrent branch of median nerve
- D. Deep branch of ulnar nerve
- E. Superficial branch of radial nerve

ANS. D

109. When falling in an outstretched hand, the most commonly dislocated carpal bone is the

- A. Scaphoid B. Lunate C. Trapezoid D. Capitate E. Hamate

ANS. B,.....fracture is to scaphoid, dislocation is to lunate.

GOOD LUCK!!!

